Lab Guide 3:

Deployment:

Q.1 Create A deployment using CLi Method using mdhack/myapache image and deployment name should be test1.

Q.2 Create A deployment using dryrun Method using mdhack/myserver image and deployment name should be test2.

Q.3 Create a deployment using image quay.io/mayank123modi/simple-webapp.

Ans: 1

```
kubectl create  deployment test1 --image mdhack/myapache

kubectl get deployments.apps

kubectl get all

kubectl get rs

kubectl get pods

kubectl describe  deployments.apps  test1

kubectl describe  rs  <rs name>

kubectl describe  pod   <pod-name>
```

```
root@master:~# kubectl create  deployment test1 --image mdhack/myapache
deployment.apps/test1 created
root@master:~# kubectl get deployments.apps
NAME    READY   UP-TO-DATE   AVAILABLE   AGE
quiz    1/1     1            1           3m51s
test1   1/1     1            1           4s
root@master:~# kubectl get all
NAME                          READY   STATUS    RESTARTS   AGE
pod/quiz-8bdf978c5-5lxzs      1/1     Running   0          3m52s
pod/test1-59bcf5dfd-6sf2l     1/1     Running   0          6s

NAME           TYPE       CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
service/quiz   NodePort   10.104.217.19   <none>        80:31086/TCP   3m37s

NAME                     READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/quiz     1/1     1            1           3m53s
deployment.apps/test1    1/1     1            1           6s

NAME                                DESIRED   CURRENT   READY   AGE
replicaset.apps/quiz-8bdf978c5      1         1         1       3m52s
replicaset.apps/test1-59bcf5dfd     1         1         1       6s
root@master:~# kubectl get rs
NAME              DESIRED   CURRENT   READY   AGE
quiz-8bdf978c5    1         1         1       3m54s
test1-59bcf5dfd   1         1         1       8s
root@master:~# kubectl get pods
NAME                     READY   STATUS    RESTARTS   AGE
quiz-8bdf978c5-5lxzs     1/1     Running   0          3m57s
test1-59bcf5dfd-6sf2l    1/1     Running   0          11s
```

```
root@master:~# kubectl describe  deployments.apps  test1
Name:                   test1
Namespace:              learning
CreationTimestamp:      Fri, 05 May 2023 05:15:27 +0000
Labels:                 app=test1
Annotations:            deployment.kubernetes.io/revision: 1
Selector:               app=test1
Replicas:               1 desired | 1 updated | 1 total | 1 available | 0 unavailable
StrategyType:           RollingUpdate
MinReadySeconds:        0
RollingUpdateStrategy:  25% max unavailable, 25% max surge
Pod Template:
  Labels:  app=test1
  Containers:
   myapache:
    Image:        mdhack/myapache
    Port:         <none>
    Host Port:    <none>
    Environment:  <none>
    Mounts:       <none>
  Volumes:        <none>
Conditions:
  Type           Status  Reason
  ----           ------  ------
  Available      True    MinimumReplicasAvailable
  Progressing    True    NewReplicaSetAvailable
OldReplicaSets:  <none>
NewReplicaSet:   test1-59bcf5dfd (1/1 replicas created)
Events:
  Type    Reason             Age    From                   Message
  ----    ------             ----   ----                   -------
  Normal  ScalingReplicaSet  2m41s  deployment-controller  Scaled up replica set test1-59bcf5dfd to 1
```

Ans: 2

kubectl create deployment test2 --image mdhack/myserver --dry-run -o yaml > dep1.yaml

cat dep1.yaml

kubectl create -f dep1.yaml

kubectl get pods

kubectl get rs

```
root@master:~# kubectl create  deployment  test2 --image mdhack/myserver --dry-run  -o yaml  > dep1.yaml
W0505 05:19:40.845882    6273 helpers.go:636] --dry-run is deprecated and can be replaced with --dry-run=client.
root@master:~# kubectl create  -f dep1.yaml
kubedeployment.apps/test2 created
root@master:~# kubectl get deployments.apps
NAME    READY   UP-TO-DATE   AVAILABLE   AGE
quiz    1/1     1            1           8m9s
test1   1/1     1            1           4m22s
test2   0/1     1            0           2s
root@master:~# kubectl get pods
NAME                     READY   STATUS    RESTARTS   AGE
quiz-8bdf978c5-5lxzs     1/1     Running   0          8m9s
test1-59bcf5dfd-6sf2l    1/1     Running   0          4m23s
test2-b449f799f-r52mh    1/1     Running   0          3s
root@master:~# kubectl get rs
NAME              DESIRED   CURRENT   READY   AGE
quiz-8bdf978c5    1         1         1       8m11s
test1-59bcf5dfd   1         1         1       4m25s
test2-b449f799f   1         1         1       5s
```

```
root@master:~# cat dep1.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  creationTimestamp: null
  labels:
    app: test2
  name: test2
spec:
  replicas: 1
  selector:
    matchLabels:
      app: test2
  strategy: {}
  template:
    metadata:
      creationTimestamp: null
      labels:
        app: test2
    spec:
      containers:
      - image: mdhack/myserver
        name: myserver
        resources: {}
status: {}
```

Ans: 3

kubectl create deployment anyname --image quay.io/mayank123modi/simple-webapp

kubectl get deployments.apps

kubectl get pods

```
root@master:~# kubectl create  deployment   anyname  --image quay.io/mayank123modi/simple-webapp
deployment.apps/anyname created
root@master:~# kubectl get deployments.apps
NAME      READY   UP-TO-DATE   AVAILABLE   AGE
anyname   0/1     1            0           5s
quiz      1/1     1            1           10m
test1     1/1     1            1           6m34s
test2     1/1     1            1           2m14s
root@master:~# kubectl get pods
NAME                       READY   STATUS    RESTARTS   AGE
anyname-bb544bf9-gvpg4     1/1     Running   0          7s
quiz-8bdf978c5-5lxzs       1/1     Running   0          10m
test1-59bcf5dfd-6sf2l      1/1     Running   0          6m36s
test2-b449f799f-r52mh      1/1     Running   0          2m16s
root@master:~#
```

Clean up your environment

```
root@master:~# kubectl delete  all --all --force
warning: Immediate deletion does not wait for confirmation that the runnin
n on the cluster indefinitely.
pod "anyname-bb544bf9-gvpg4" force deleted
pod "quiz-8bdf978c5-5lxzs" force deleted
pod "test1-59bcf5dfd-6sf2l" force deleted
pod "test2-b449f799f-r52mh" force deleted
service "quiz" force deleted
deployment.apps "anyname" force deleted
deployment.apps "quiz" force deleted
deployment.apps "test1" force deleted
deployment.apps "test2" force deleted
replicaset.apps "test2-b449f799f" force deleted
root@master:~#
```